

EE 4301 Capstone Project

Topic: Microstrip Patch Antenna at 2.4 GHz

I. Introduction

Microstrip patch antennas are a class of planar antennas widely used in modern wireless communication systems. They consist of a thin metallic radiating patch printed on a dielectric substrate, with a conductive ground plane on the opposite side. Due to their low profile, compact size, and compatibility with printed circuit board (PCB) fabrication techniques, microstrip patch antennas are commonly integrated into wireless devices such as Wi-Fi routers, Bluetooth modules, smartphones, and Internet-of-Things (IoT) systems [1][2].

At microwave frequencies, particularly in the 2.4 GHz industrial, scientific, and medical (ISM) band, microstrip patch antennas provide a practical balance between performance and manufacturability. Their planar geometry allows easy integration with RF front-end circuitry, while their resonant behavior enables predictable radiation characteristics. By adjusting the patch dimensions and substrate properties, designers can precisely control the antenna's operating frequency, impedance matching, and radiation pattern [2].

I chose this topic because of its strong connection between electromagnetic theory and real-world engineering applications. In Spring 2026, I will be working as an RF engineering co-op intern, where antennas, wireless propagation, and RF system behavior will be central to my responsibilities. Understanding how microstrip patch antennas are designed from first principles helps bridge the gap between classroom electromagnetics and practical RF engineering workflows. This project provides an opportunity to apply theoretical concepts such as wave propagation in dielectrics, boundary conditions, and resonance to a realistic antenna design problem.

II. Applicable Theory

Microstrip patch antennas operate based on fundamental electromagnetic principles governed by Maxwell's equations, particularly the behavior of time-varying electric and magnetic fields in bounded dielectric structures. A microstrip patch antenna consists of a thin metallic radiating patch printed on a dielectric substrate and backed by a conductive ground plane, forming a planar resonant system. When excited by a radio-frequency signal, electromagnetic waves propagate within this layered medium and establish standing wave patterns between the patch and the ground plane. Radiation occurs primarily at the open edges of the patch, where electromagnetic boundary conditions are not fully satisfied and electric field lines extend into free space, producing fringing fields that act as the primary radiation mechanism.

A key electromagnetic concept involved in microstrip patch antenna operation is wave propagation in dielectric media. Unlike antennas operating entirely in free space, the electromagnetic fields in a microstrip structure exist partly within the dielectric substrate and partly in air. This mixed propagation environment alters the phase velocity and wavelength of the guided waves. To account for this behavior, an effective dielectric constant is introduced, which models the combined influence of the substrate and surrounding air on wave propagation. The effective permittivity directly determines the guided wavelength and strongly influences the antenna's resonant dimensions, size, and bandwidth. Consequently, substrate parameters such as relative permittivity and thickness play a critical role in antenna performance.

Electromagnetic resonance governs the operating frequency of the antenna. Rectangular microstrip patch antennas most commonly operate in the dominant TM_{10} mode, in which the electric field varies sinusoidally along the length of the patch while remaining approximately uniform along the width. In this mode, the effective resonant length of the patch is approximately equal to one-half of the guided wavelength. This behavior is often described using the cavity model approximation, which treats the patch as a resonant cavity with electric walls at the patch and ground plane and magnetic walls along the radiating edges. This model provides physical insight into standing-wave formation and resonance conditions.

Boundary conditions and fringing effects significantly influence practical antenna design. At the radiating edges of the patch, the electric field does not terminate abruptly but instead extends beyond the physical conductor into the surrounding air. These fringing fields effectively increase the electrical length of the antenna relative to its physical dimensions. As a result, a length correction factor must be applied during the design process to ensure that the antenna resonates at the desired frequency. This effect highlights the direct impact of electromagnetic boundary conditions on real-world antenna geometry.

Transmission line theory is applied to analyze antenna excitation and impedance matching. Microstrip transmission lines guide electromagnetic energy from the source to the patch, and the input impedance of the patch varies along its length. At the radiating edge, the impedance is relatively high, while it decreases toward the center of the patch. By employing an inset feed, the feed point can be positioned such that the antenna input impedance matches a standard 50-ohm transmission line. This matching minimizes reflections and maximizes power transfer between the source and the antenna.

Finally, the antenna's radiation behavior and loss mechanisms reflect the conversion of guided electromagnetic energy into radiated energy. A single rectangular microstrip patch antenna produces a broadside radiation pattern with moderate gain and inherently narrow bandwidth. These effects illustrate fundamental trade-offs between antenna size, efficiency, bandwidth, and manufacturability.

Together, these electromagnetic topics: Maxwell's equations, wave propagation in dielectrics, resonance, boundary conditions, transmission line theory, and radiation all form the theoretical foundation for the microstrip patch antenna design.

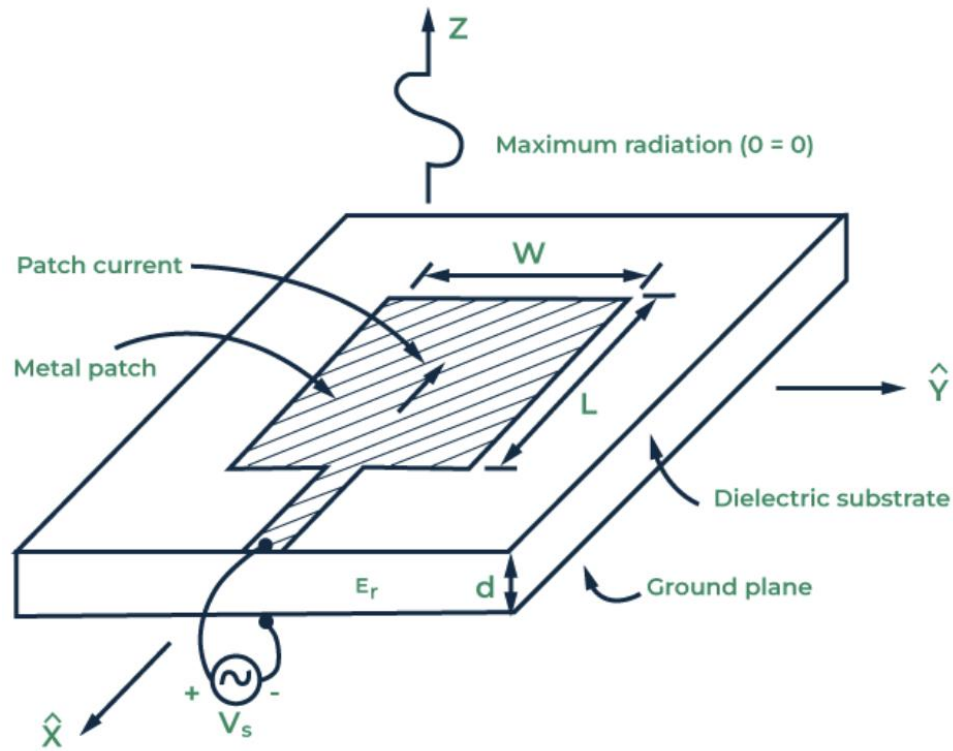


Figure 2.1: Top View of Rectangular Microstrip Antenna

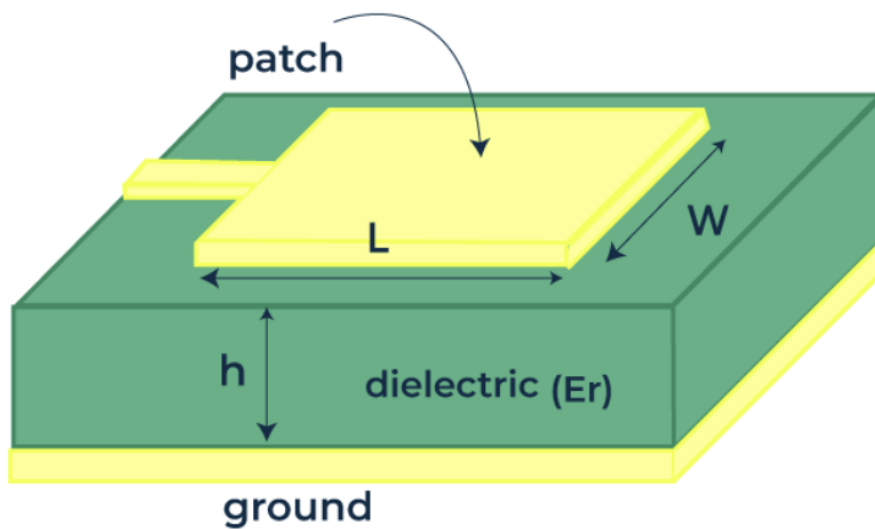


Figure 2.2: Side View of Rectangular Microstrip Antenna

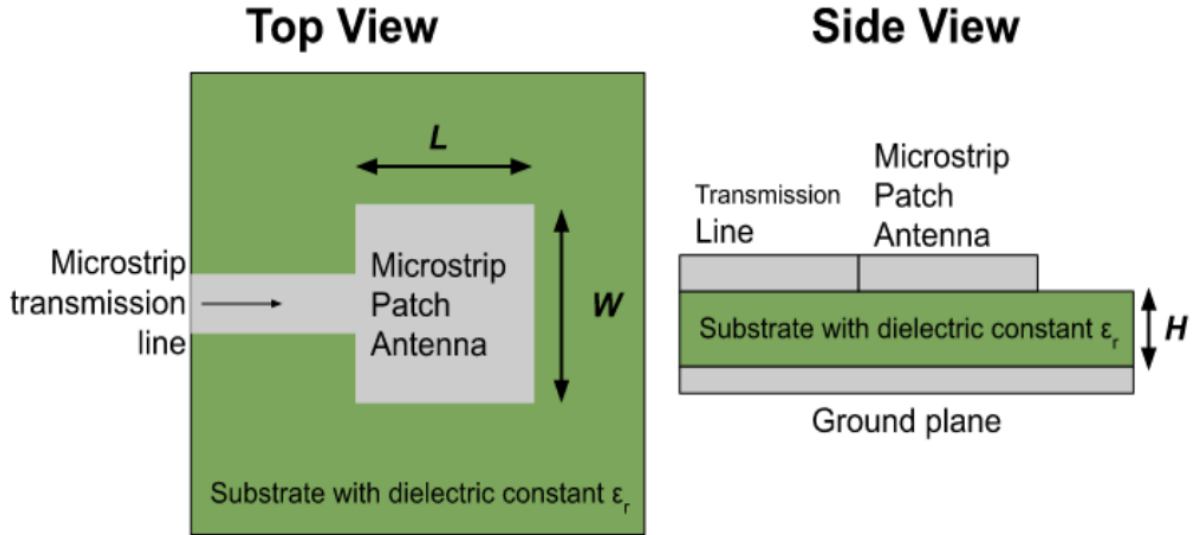


Figure 2.3: Combined View from Cadance (PCB Style)

III. Example Design Problem – Patch Antenna at 2.4 GHz

The objective of this design example is to create a rectangular microstrip patch antenna operating at 2.40 GHz using an FR-4 substrate with relative permittivity $\epsilon_r = 4.4$, substrate thickness $h = 1.6\text{mm}$, and an inset-fed 50-ohm microstrip line.

Step 1: Patch Width

The patch width is selected to improve radiation efficiency and bandwidth:

$$W = \frac{c}{2f} \sqrt{\frac{2}{\epsilon_r + 1}}$$

Substituting the design values:

$$W \approx 38 \text{ mm}$$

Step 2: Effective Dielectric Constant

Because fields exist partly in air and partly in the substrate, the effective dielectric constant is calculated as:

$$\epsilon_{\text{eff}} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left(1 + \frac{12h}{W}\right)^{-1/2}$$

$$\epsilon_{\text{eff}} \approx 4.09$$

Step 3: Length Extension Due to Fringing Fields

$$\Delta L = 0.412h \frac{(\epsilon_{\text{eff}} + 0.3)(W/h + 0.264)}{(\epsilon_{\text{eff}} - 0.258)(W/h + 0.8)}$$
$$\Delta L \approx 0.74 \text{ mm}$$

Step 4: Effective Resonant Length

$$L_{\text{eff}} = \frac{c}{2f\sqrt{\epsilon_{\text{eff}}}}$$
$$L_{\text{eff}} \approx 30.9 \text{ mm}$$

Step 5: Physical Patch Length

$$L = L_{\text{eff}} - 2\Delta L$$
$$L \approx 29.4 \text{ mm}$$

Step 6: Inset Feed Position for Impedance Matching

Assuming an edge impedance of approximately 300Ω , the inset feed position is determined using:

$$R_{\text{in}}(y_0) = R_{\text{edge}} \cos^2\left(\frac{\pi y_0}{L}\right)$$

Solving for a 50-ohm match yields:

$$y_0 \approx 10.9 \text{ mm}$$

This specifies how far the feed line must extend into the patch to achieve proper impedance matching.

Final Antenna Design Summary

The designed antenna is expected to exhibit a gain of approximately 4–6 dB, a bandwidth of 2–3% (50–75 MHz), linear polarization, and a broadside radiation pattern. Due to dielectric losses in FR-4, the radiation efficiency is expected to be approximately 50–60%, which is typical for PCB-based microstrip patch antennas.

Parameter	Value
Patch Width (W)	38 mm
Patch Length (L)	29.4 mm
Substrate Height (h)	1.6 mm
Effective Permittivity (ϵ_{eff})	4.09
Length Extension (ΔL)	0.74 mm
Inset Feed Position (y_0)	10.9 mm

IV. Conclusion

Designing a 2.4 GHz microstrip patch antenna provided a practical opportunity to apply core electromagnetic theory to a realistic engineering problem. Using the transmission-line model and cavity approximation, I analyzed how substrate properties, physical dimensions, and fringing electric fields govern the resonant behavior of the antenna. This project reinforced key electromagnetic concepts such as wave propagation in dielectric media, boundary conditions, effective permittivity, and radiation due to field discontinuities.

The numerical design example followed the systematic workflow commonly used in RF and antenna engineering. Starting from system-level requirements, I determined the patch width, effective dielectric constant, resonant length, and inset feed position, resulting in a complete antenna design for operation at 2.4 GHz on an FR-4 substrate. These calculations highlighted the sensitivity of antenna performance to substrate thickness and dielectric constant, as well as the trade-offs between antenna size, efficiency, and bandwidth. The inset feed design demonstrated how impedance matching can be achieved analytically rather than through trial-and-error.

This project deepened my understanding of how electromagnetic theory translates into physical antenna structures, knowledge that will be directly applicable during my upcoming RF engineering internship. It also emphasized the importance of accurate modeling, parameter selection, and consideration of material losses when working with low-cost substrates such as FR-4.

Future extensions of this work could include designing dual-band or circularly polarized patch antennas, optimizing the design using full-wave electromagnetic simulation tools such as HFSS or CST, or developing a patch antenna array to increase gain and directivity. Overall, this project demonstrated the practical value of electromagnetics in real-world engineering applications and strengthened my analytical and design skills in RF and antenna engineering.

V. References

- [1] F. T. Ulaby and U. Ravaioli, *Fundamentals of Applied Electromagnetics*, 8th ed., Pearson, 2020.
- [2] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley, 2016.
- [3] D. M. Pozar, *Microwave Engineering*, 4th ed., Wiley, 2012.
- [4] “Microstrip Patch Antenna,” Antenna-Theory.com. [Online]. Available: <https://www.antenna-theory.com>. Accessed: Sept. 2025.
- [5] GeeksforGeeks, “Microstrip Patch Antenna,” 2023. [Online]. Available: <https://www.geeksforgeeks.org/electronics-engineering/microstrip-patch-antenna/>. Accessed: Sept. 2025.
- [6] Cadence PCB Solutions, “Microstrip Patch Antenna Introduction,” 2023. [Online]. Available: <https://resources.pcb.cadence.com/blog/2023-microstrip-path-antenna-introduction>. Accessed: Sept. 2025.